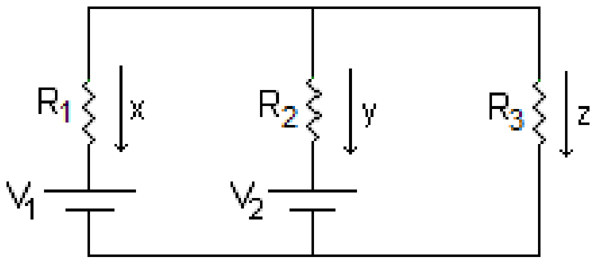
Physics 4B Lab 7: *The Multiloop Circuit*

The objective of this lab exercise is to predict, measure and analyze the behavior of a *multiloop* circuit.

Our standard "three branch" multiloop circuit consists of two batteries and three resistors:



We will construct this circuit for Lab 7 using the three resistors from Lab 5. The breadboard has two built-in power supplies that we will use in place of the two batteries. **We will use a variable power supply for V_1 and a fixed 5.00 V power supply for V_2.**

We will vary the value of V_1 and observe the corresponding value of the current in the third branch, i.e. current "z". We can then use the collected data to determine whether or not the circuit behaves as expected.

How do we expect the circuit to behave? We can treat this as a standard multiloop circuit problem, i.e. we can write the three necessary equations and solve. But what do we solve for? Since we will measure current "z", we will graph current "z" on our y-axis, we want to derive an equation for current "z" in terms of the elements in the circuit, i.e. R_1 R_2 R_3 V_1 and V_2. Note that all of these, with the exception of V_1, will be constants in this experiment.

Our three equations…

Currents: $$x + y + z = 0$$

Big loop: $$V_1 + x R_1 - z R_3 = 0 \quad or \quad x = \frac{z R_3}{R_1} - \frac{V_1}{R_1}$$

Right loop: $$V_2 + y R_2 - z R_3 = 0 \quad or \quad y = \frac{z R_3}{R_2} - \frac{V_2}{R_2}$$

Note: we chose the "big loop" and "right loop" because we want to solve for current z, so all three equations must include "z".

We can now eliminate x and y from these three equations, by subbing the latter two equations into the first, and solve for current "z":

$$\frac{z\,R_3}{R_1} - \frac{V_1}{R_1} + \frac{z\,R_3}{R_2} - \frac{V_2}{R_2} + z = 0$$

Or:

$$\frac{z\,R_3}{R_1} + \frac{z\,R_3}{R_2} + \frac{z\,R_3}{R_3} = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

Or:

$$z\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) = \frac{V_1}{R_1 R_3} + \frac{V_2}{R_2 R_3}$$

Or:

$$z\left(\frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 R_2 R_3}\right) = \frac{V_1}{R_1 R_3} + \frac{V_2}{R_2 R_3}$$

Or:

$$z\,(R_1 R_2 + R_1 R_3 + R_2 R_3) = R_2\,V_1 + R_1\,V_2$$

Or:

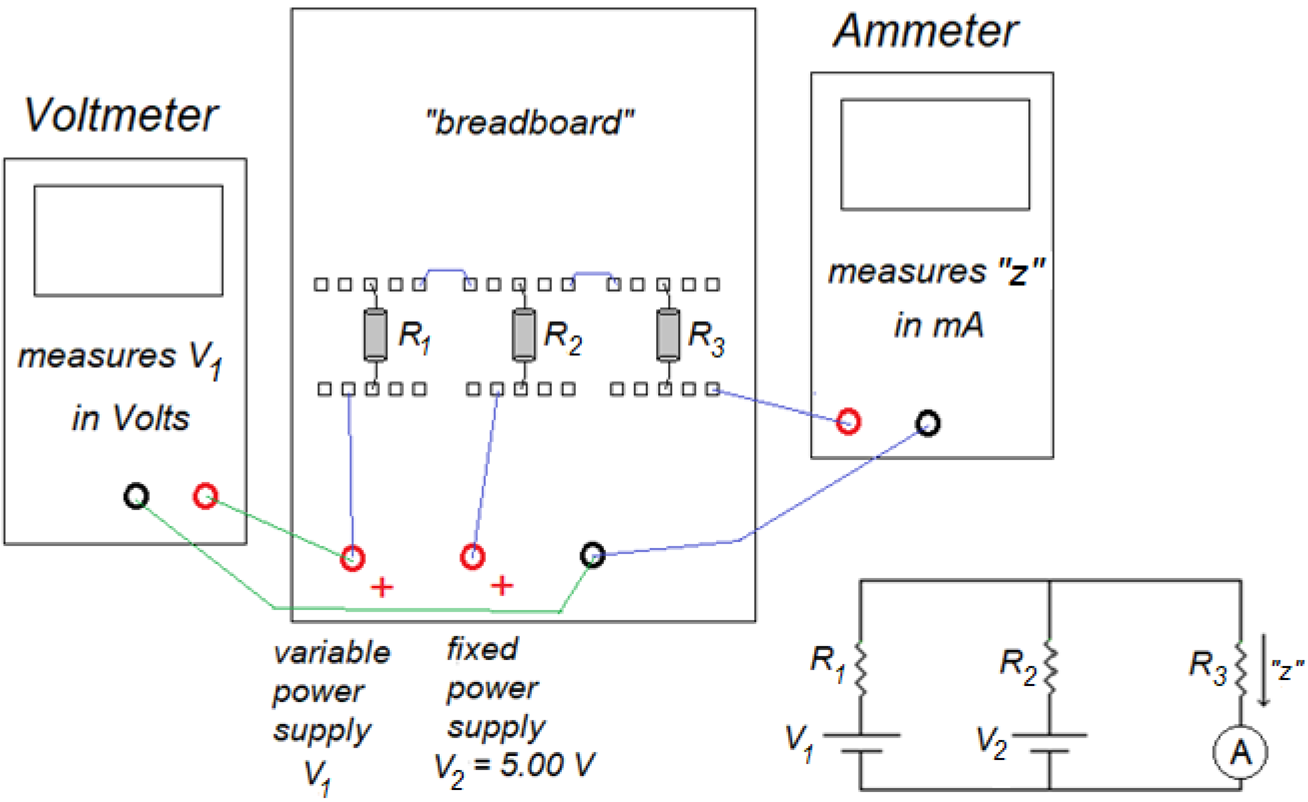
$$z = \left(\frac{R_2}{\alpha}\right) V_1 + \left(\frac{R_1\,V_2}{\alpha}\right)$$

$$where \quad \alpha = R_1 R_2 + R_1 R_3 + R_2 R_3$$

Notice that everything in parentheses in the above equation will be constant during our experiment. (Recall that, from the explanation above, **V_2 = 5.00 Volts**.) Which means that if we collect data of values for V_1 and the corresponding current "z", we can graph *z* vs *V_1* and expect a straight line with:

$$slope\ of\ \frac{R_2}{\alpha} \qquad and \qquad y - intercept\ of\ \frac{R_1\,V_2}{\alpha}$$

The apparatus for Lab 7 is very similar to that of Labs 5 and 6:



Procedure

The very first thing you will need to fill in on your Excel sheet (after your name and date, of course…) is the value of the resistance of your three resistors. These values come from your results of Lab 5.

Be sure to use the values in kΩ, i.e. the values directly from your graph.

There are two trials for Lab 7. They are identical in procedure; the only difference between the two trials is the order of the resistors, so the values of R_1 R_2 and R_3 are different from Trial 1 to Trial 2 (although the same three resistors are used for each trial.)

- Fill in the first data table for Trial 1. ***Link the entries for "Orig R" to the table above.***

  This table serves two purposes: to establish the values for R_1 R_2 and R_3 for Trial 1, and to adjust the value of R_3. Since the ammeter is part of the third branch, its resistance must be included as part of the resistance on the third branch. The ammeter has a small resistance of 0.013 kΩ (i.e. 13 Ohms), and it is in series with R_3. So the "adjusted" value of R_3 is simply the original value plus 0.013 kΩ. (The "adjusted" values of R_1 and R_2 are the same as the original values.)

- For values of V_1 of approximately 1, 2, 3,… 10 Volts (just as we did for our procedure in Lab 5), record the corresponding values of current "z". Then create a graph of *z* vs V_1. Add the best-fit line and equation. Format the equation to three decimal places.

- Complete the calculations section:

    - Calculate the value of α. All resistances are in kΩ, so the units for α will be $(\text{k}\Omega)^2$.

    - Calculate the expected value of the slope of the graph, with appropriate units.

    - Calculate the expected value of the y-intercept, with appropriate units.

    - Calculate the percent difference for each.

You should find that the measured slope and intercept are close to the expected, within a few percent and possibly within 1%.

Repeat the above procedure for Trial 2. Note that the order of the resistors is changed. The procedure is exactly the same, but the results will be different because of the different values of R_1 R_2 and R_3. The expected values for the slope and intercept in Trial 2 will be different than for Trial 1, but should match to the measured values for Trial 2.